

Spatial diagonal weight $W = \text{diag}(w_1, \dots, w_N)$, $0 < w_0 \leq w_i \leq w_1$

solution-preserving;
stability constant unchanged

frozen spectrum: directions
unreachable, scale bounded

the only route into
the one-step update

Channel S | zeroth-order: solution / fit

zero-residual set, global minimizer,
residual-error bridge;
independent of W for $w > 0$

**Level R | frozen linearization
(not an independent channel)**

$H_{\text{lin}} = J^{\top} W J$ and its spectrum;
acts only indirectly via
multi-step gradient evolution

Channel θ | first-order: algorithm

$g_w = J^{\top} W r$, iteration trajectory,
stationary points and
basins of attraction

Level R acts on channel θ only indirectly: within frozen training windows, via multi-step g_w evolution (dashed arrows)

Controlled remainder E_{rem} (not a fourth channel)

residual curvature $H_{\text{nl}} = \sum_i w_i r_i \nabla^2 r_i$, 3rd+-order Taylor terms, in-window curvature drift,
preconditioner variation, exogenous jumps
—— only explicit, controlled upper bounds